Shashank Kumar

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Research Interests

Making the behaviour of autonomous agents checkable by parties other than the operator: whether an agent stayed in scope, stopped on time and destroyed its access, and, when containment fails, what the receiving platform can establish from its own records. Recent work analyses real agent data, including 838,000 posts by AI agents on Moltbook and the two 2026 containment incidents, and builds the tools needed to test claims against it. I state what the evidence does and does not support, and I publish corrections to my own work.

Selected Analysis

Attestation Gaps – Why Signature-Based Attribution Fails on Agent Escapes Sep 2026
Apart Research, AI Incident Response Sprint (Containment track) • apartresearch.com/…/kairos-2vka

- Implemented per-action signing for agent actions and tested it against workloads shaped like the 2026 German-wiki and Hugging Face incidents. It attributed none of the escaped actions, and partial signing produced a misleading 100% compliance rate.
- **The inverse works:** unsigned traffic from a lab's declared address ranges is evidence a platform can check from its own logs (15,000 of 15,000 synthetic agent edits flagged, no false positives; real-log validation still open).
- Audited OpenAI's published GPTBot and SearchBot range files: 251 and 315 days stale, 14.6% of addresses shared between products, and none of 60 prefixes reliable enough to support a finding. Turned this into four checkable registry requirements for regulators. A reviewer independently reproduced the audit.

Swarmchaser – Coordinated AI-Agent Swarms on Moltbook Oct 2026
AI Swarm Dynamics Hackathon (AI Village × Grove Research) • sidthebuilder.github.io/swarmchaser • code

- Analysed 838,000 posts and 172,000 comments; found 119 swarms covering 8.7% of active agents. None of the 91 swarms with three or more known owners is run by a single owner, so operator-level attribution mostly fails.
- Found 18 swarms steered by a statistically significant outside link or skill file (one-sided Fisher exact test, Bonferroni-corrected, carried by at least two owners).
- **Checked against a placebo:** an apparent 4.9-fold agent-to-agent contagion effect was about half self-selection, and only 0.11% of 107,278 campaign members saw a campaign post before joining.
- Early warning by replay: all 30 top swarms flagged at six-hour checkpoints, with 97.5% of their posts after the first flag. Validated with planted swarms (6 of 7 recovered exactly) and a hand-labelled precision estimate (65%, or 77% excluding unclear cases). 108 tests.

Provenant – Graph-Backed Agents on Real Agent Data Oct 2026
WeMakeDevs × FalkorDB Graph Hacks • github.com/sidthebuilder/Provenant

- An investigation agent that replays a proposed block on a copy of the graph before recommending it. On Moltbook, blocking one skill file stops 172 of a swarm's 207 posts; blocking the top PageRank link would hit 624 ordinary posts to stop 22.
- Shared-memory benchmark on 1,020 real moments when AI Village agents searched their own history: 73% of the messages they needed were written by another agent; graph memory recovered 52% in its top 20, against 42% for recent messages (held-out half).

Research Projects

CHRONOS – Proof-Carrying Containment for Ephemeral Agents Rust, arkworks, TFHE-rs

- Composes fully homomorphic encryption, a Wesolowski verifiable delay function, a Groth16 argument and a machine-checked containment monitor, with no trusted hardware. A single 128-byte proof attests that the agent held a key released only by sequential work, destroyed it, and stayed within its capability limits.
- **Containment as invariants:** seven order-theoretic invariants (for example: capabilities only shrink, shutdown is reachable from every state, no action above a set uncertainty threshold), all checked over 186,624 abstract states before the agent accepts a request.
- **Measured, not asserted:** erasure circuit of 12,966 R1CS constraints; proofs verify in 1 to 2 ms; VDF throughput flat across T at about 480,000 squarings per second; in-circuit authenticated decryption at 1,204 constraints. 336 passing tests.
- **Limits, stated plainly:** the trusted-setup ceremony is implemented but has never been run; erasure rests on an operating-system memory assumption; the uncertainty bound holds only if the agent reports honestly.
- github.com/ChronosResearch/project-chronos (Apache-2.0)

PAAC – Formal Verification for Self-Modifying Code Python, Z3 SMT

- Checks agent-proposed code changes against safety axioms with Z3 bounded model checking *before* execution. 398 tests pass; two confirmed composition vulnerabilities (accepted changes that combine into a violation) remain open. github.com/ChronosResearch/paac

Other Python

- Epistemic infrastructure for retrieval systems: replayable reasoning traces that separate parsing, grounding and reasoning failures in RAG. CellHawk: GPS-denied navigation from cellular signal strength with an Extended Kalman Filter (simulation only, no hardware testing).

Research Practice and Self-Correction

The clearest evidence of how I work is the record of errors I found in my own published work and disclosed.

- Found that an earlier CHRONOS erasure circuit only compared a prover-supplied buffer with a constant, so a prover who had never seen the key could pass; the central claim was vacuous. Documented it and rebuilt the circuit (8,267 constraints in v4, 12,966 in v4.1).
- Corrected a published constraint count of about 180,000, of which about 160,000 were filler, and withdrew per-component figures that had been estimated rather than measured.
- Traced non-monotonic VDF timings to a trial division in the Fiat–Shamir step rather than the sequential squaring being measured.
- Found the hard-coded "RSA-2048" modulus had 618 digits and was divisible by 5, which would have removed the sequentiality the construction rests on.
- Removed three claimed contributions the code did not deliver, rather than leaving them unexplained.
- In PAAC, found that three of five safety axioms never fired, so unchecked code was reported safe; fixed it and corrected a test count that had never been measured.
- Every defect is catalogued publicly in each repository's audit file.

Preprints (not peer-reviewed)

- Project CHRONOS: A Fully Homomorphic Ephemeral AI Agent. SSRN 6950898, Jun 2026 (latest revision v4.1, Aug 2026).
- The Provably Aligned AI Core: A Deterministic Safety Wrapper. SSRN 6879218, Jun 2026.
- The Zero-Training Enterprise Brain: A Live Semantic Reasoning Architecture. SSRN 6842778, May 2026.
- Building Epistemic Infrastructure for Enterprise AI Search. SSRN 6585978, Apr 2026.
- Beyond Single Prompts: Engineering a Multi-Agent Workgroup for End-to-End Tabular Data Science. Technical report, Mar 2026.

Experience

AI Systems Evaluator and Red Teamer, Alignerr (remote) Apr 2025 – present

- Rated top 5% (platform-issued badge) across 1,000+ adversarial evaluation tasks. Wrote 500+ red-team prompts against pre-release frontier models, targeting jailbreak resistance, factual grounding and instruction-following.

Full Stack AI Engineer, BitNorm (Netherlands, remote) Nov 2025 – present

- Security-audited a pre-launch Rust and GraphQL platform; found critical issues in payment handling, authentication and secrets management, including a PCI-DSS violation. Delivered a prioritised remediation plan the engineering team accepted.

Community Member, Cohere Open Science Community (remote) Sep 2026 – present

- Admitted after a technical review of open-source verifiable-AI projects.

Freelance AI and Full-Stack Developer, Upwork and Fiverr (remote) Mar 2021 – present

- 30+ production systems across fintech, healthcare and SaaS; sustained 4.8/5 client rating; Top Rated on Upwork. Annotated 10,000+ multilingual text and voice samples for NLP and speech-recognition datasets; evaluation work via Outlier AI (2024 – present).

Skills

- **Analysis:** observational data analysis, negative-control (placebo) checks, Fisher exact tests with multiple-comparison correction, held-out evaluation, planted-signal recall tests
- **Cryptography and formal methods:** Groth16 (arkworks), Poseidon, verifiable delay functions, BLS12-381, ML-DSA, authenticated encryption, Z3 SMT, invariant and lattice-based state modelling
- **Languages and systems:** Python, Rust, TypeScript, C++, C#, SQL; FastAPI, GraphQL, PostgreSQL, Redis, FalkorDB, Docker, GitHub Actions, AWS, Azure
- **AI/ML:** adversarial red-teaming, multi-agent systems, RAG, FHE inference (TFHE-rs), LoRA and DPO fine-tuning

Education

B.Sc. Computer Science, University of the People Apr 2025 – expected 2027

B.A. (English, History, Hindi), Swami Vivekanand Subharti University, Meerut 2021 – 2025

Self-taught in cryptography and formal methods. CS50: Introduction to Computer Science (Harvard, edX, 2024).